Climate-Conditioned Robust Portfolios in Latin American Equity Markets

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Abstract

Latin American economies face a twofold climate vulnerability: high exposure to physical risk, modulated by the ENSO cycle, and transition risk arising from the concentration of their equity markets in extractive and carbon-intensive sectors. This paper proposes a framework to test whether climate information improves the control of tail risk in distributionally robust equity portfolios in four regional markets: Brazil (IBOVESPA), Mexico (IPC), Colombia (COLCAP), and Chile (IPSA). The framework uses a Kalman-filtered climate state (considering carbon prices, regional temperatures, El Niño-Southern Oscillation (ENSO), and the dominant commodity) to reweight historical return scenarios and determine the Wasserstein transport cost. The exact worst-case CVaR is solved as a convex program under EU Paris-aligned constraints with a cross-validated radius and is evaluated out of sample against nested benchmarks with bootstrap inference. The climate-conditioned model is statistically indistinguishable from its climate-free counterpart, and no market-level difference survives multiple testing adjustments. Transition-aligned portfolios can be built with tail risk comparable to country ETFs over one ENSO cycle and with transition intensities. There is a closed-form climate-conditioned ambiguity set for Latin American equities. Equity responses to ENSO news are market-wide and episodic, limiting the value of firm-level physical exposure constraints.

Keywords: Climate uncertainty, distributionally robust optimization, ENSO, Emerging markets.
JEL Codes: G11, G15, Q54, C61, C32.

Resumen

Las economías latinoamericanas se enfrentan a una doble vulnerabilidad climática: una elevada exposición al riesgo físico, modulada por el ciclo ENSO, y el riesgo de transición derivado de la concentración de sus mercados de valores en sectores extractivos y con altas emisiones de carbono. Este artículo propone un marco para comprobar si la información climática mejora el control del riesgo de cola en carteras de acciones robustas desde el punto de vista distributivo en cuatro mercados regionales: Brasil (IBOVESPA), México (IPC), Colombia (COLCAP) y Chile (IPSA). El enfoque utiliza un estado climático filtrado por Kalman (que tiene en cuenta los precios del carbono, las temperaturas regionales, El Niño-Oscilación del Sur (ENSO) y la materia prima dominante) para reponderar los escenarios de rentabilidad históricos y determinar el coste de Wasserstein. El CVaR exacto del peor caso se

resuelve como un programa convexo bajo restricciones de la UE alineadas con el Acuerdo de París, con un radio validado de forma cruzada, y se evalúa fuera de la muestra frente a índices de referencia anidados mediante inferencia bootstrap. El modelo condicionado por el clima es estadísticamente indistinguible de su homólogo sin condicionamiento climático y ninguna diferencia a nivel de mercado resiste los ajustes por pruebas múltiples. Se pueden construir portafolios alineados con la transición con un riesgo de cola comparable al de los ETF por países a lo largo de un ciclo de ENSO y con intensidades de transición. Existe un conjunto de ambigüedad condicionado al clima en forma cerrada para la renta variable latinoamericana. Las respuestas de la renta variable a las noticias sobre ENSO son generalizadas en todo el mercado y episódicas, lo que limita el valor de las restricciones de exposición física a nivel de empresa.

Palabras clave: Incertidumbre climática, optimización distribucionalmente robusta, ENSO, mercados emergentes.
Clasificación JEL: G11, G15, Q54, C61, C32.

1. Introduction

Exposure to climate risks, both physical and transitional, is one of the central challenges for Latin American equity markets. The region's susceptibility to extreme climate events, coupled with its reliance on carbon-intensive sectors, creates a vulnerability that differs significantly from that observed in advanced economies. The El Niño-Southern Oscillation (ENSO) modulates precipitation, river discharge, and hydropower output through the physical channel across the region (Poveda et al., 2001; Cai et al., 2020). This has been documented for Colombian hydropower plants (Pulgarín-Morales and Krueger, 2025) and the central Chile megadrought (Garreaud et al., 2020). The macroeconomic effects of ENSO are also persistent (Cashin et al., 2017; Callahan and Mankin, 2023). Through the transition channel, local indices concentrate in extractive and carbon-intensive sectors, exposing financial institutions and portfolios to transition risk (Ramírez et al., 2020). Carbon pricing in the four markets is heterogeneous. Mexico introduced a carbon tax in 2014, while Colombia and Chile introduced one in 2017. Brazil had no binding carbon price during the sample period.

International literature has documented the pricing of climate risks. For example, Bolton and Kacperczyk (2021, 2023) identified a carbon-risk premium in a global sample that included emerging markets. Pastor et al. (2021) demonstrated that sustainable preferences alter equilibrium returns. Engle et al. (2020) revealed that climate news risk could be hedged with equity portfolios. Bressan et al. (2024) discovered significant heterogeneity within sectors regarding the impact of physical shocks on firms' cash flows. Additionally, Roncoroni et al. (2021) showed that climate shocks propagate asymmetrically through the region's financial networks. However, regional evidence on risk-adjusted performance remains scarce (Pesce et al., 2024; Monzon Citalan et al., 2025), and the incorporation of climate information into quantitative portfolio decisions in Latin America is limited.

The integration of climate risk into asset valuation has rested largely on Integrated Assessment Models (IAMs), whose deterministic architecture tends to underestimate the

convexity of geophysical damage and the weight of the tails of the catastrophe distribution (Weitzman, 2009; Keen et al., 2022). In settings of high regulatory volatility, where climate-policy trajectories can shift with electoral cycles, investors face deep uncertainty, or ambiguity in the Knightian sense, in addition to risk with a known distribution. To address this problem, distributionally robust optimization (DRO) with ambiguity sets under the Wasserstein metric offers a suitable conceptual framework (Esfahani & Kuhn, 2018; Blanchet & Murthy, 2019; Barnett et al., 2020): it optimizes the portfolio against the most adverse distribution within a ball of radius ε around the empirical distribution, without imposing restrictive parametric assumptions ex ante. Two features of the standard formulation limit its usefulness for climate-aware investing. First, the nominal distribution and the transport cost ignore the climate state, so climate information can only affect the size of the ambiguity set. Second, when the radius is large relative to return dispersion and the transport cost is Euclidean, the robust portfolio converges to the equally weighted or minimum-norm feasible portfolio (Pflug et al., 2012), so the model reduces to naive diversification.

This paper addresses both points. Using a linear Gaussian state-space model estimated by the expectation-maximization algorithm (Shumway and Stoffer, 1982) and filtered with the Kalman filter (Kalman, 1960; Durbin and Koopman, 2012), we extract climate information from four dimensions: the price of carbon, regional temperature anomalies, ENSO, and the price of each country's dominant commodity. Climate information enters the ambiguity set in two ways: (i) A kernel reweights historical return scenarios according to their climate similarity with the current state, in the spirit of Bertsimas and Kallus (2020), and (ii) the covariance of returns induced by climate news is incorporated into the transport cost so that the ambiguity ball elongates in the directions in which climate can jointly move returns. The radius is selected by rolling-origin cross-validation. The portfolios satisfy the transition limits of the EU Paris-aligned benchmarks (European Commission, 2020) as well as a limit on a data-based measure of physical exposure.

The paper makes three contributions. First, it derives the exact worst-case CVaR over a type-2 Wasserstein ball with a climate-dependent metric: the robust objective equals the nominal CVaR plus $\varepsilon\alpha^{-1/2}$ times the portfolio volatility under the climate-augmented metric. This gives the radius a dimensionless interpretation and yields a second-order cone program. Second, the model is evaluated out of sample from 2018 to 2025 in Brazil, Mexico, Colombia, and Chile. This evaluation is done against nested benchmarks that isolate the contributions of the objective, robustness, metric, and climate information. Transaction costs and bootstrap inference are also considered. Third, it measures firm-level physical exposure from data and documents that equity responses to ENSO news are mostly market-wide and episodic. The main empirical result is a precise null hypothesis: the climate-conditioned model does not differ significantly from its climate-free counterpart. The data also show that the effect of climate information on monthly CVaR is about half a percentage point in either direction.

The remainder of the paper is organized as follows. Section 2 reviews the literature and the region's institutional context. Section 3 formalizes the architecture of the DRO-KF model. Section 4 presents empirical application and results. Section 5 sets out the main conclusions and their implications for portfolio managers and fiduciary institutions.

2. Related Literature and Institutional Context

2.1 Climate Risk in Latin America: Evidence and Sectoral Exposure

Research on the financial effects of climate risk in Latin America has grown in recent years, driven by the sectoral concentration of the regional equity indices and by the greater frequency and intensity of extreme hydrometeorological events. At the aggregate level, Ramírez et al. (2020) document how Latin American financial institutions assess transition risks and the exposure of their portfolios to carbon-intensive sectors. Roncoroni et al. (2021) show that the propagation of climate shocks through the financial system (banks and investment funds) is asymmetric across the region. Chile and Colombia have high physical-risk exposure due to their heavy dependence on water resources. Additionally, at the level of individual assets, Bressan et al. (2024) find heterogeneity within sectors in how physical shocks affect firms' cash flows. Regarding valuation, Bolton & Kacperczyk (2021, 2023) provide evidence of a carbon-risk premium that penalizes firms with higher emission intensity. Reinders et al. (2023) also confirms this relationship for emerging markets.

The region's physical risk is closely tied to the ENSO cycle, whose interannual variability affects the four economies simultaneously but asymmetrically. Poveda et al. (2001) document the seasonality of ENSO effects on precipitation and river discharge in Colombia; Cai et al. (2020) review ENSO impacts on South American rainfall, agriculture and hydrology; Pulgarín-Morales and Krueger (2025) quantify ENSO effects on four Colombian hydropower plants; and Garreaud et al. (2020) analyze the central Chile megadrought. At the macroeconomic level, El Niño shocks have persistent effects on output and commodity prices (Cashin et al., 2017; Callahan and Mankin, 2023). In financial markets, Hong et al. (2019) show that stock prices underreact to drought information, which motivates measuring the response to climate news with lags. Because climate projections indicate changes in ENSO-related extremes under warming (Cai et al., 2020; IPCC, 2022), these effects are a natural candidate state variable for portfolio decisions in the region.

2.2 Carbon Regulation and Transition Risk

The four markets operate under different carbon pricing regimes, which generate different transition risk profiles. Mexico adopted a federal carbon tax in January 2014 at approximately USD 3-4 per ton of CO_2e (World Bank, 2025), and ran a pilot emissions trading system from 2020 to 2022 whose transition to an operational phase was still pending at the end of the sample. Colombia introduced its national carbon tax in 2017 (Law 1819 of 2016), at about USD 5 per tCO2e, with a mechanism that allows firms to offset the tax with certified emission reductions. Chile's green tax on large stationary sources (Law 20,780) has applied since 2017 at USD 5 per ton of CO_2e, and the Framework Law on Climate Change (Law 21,455 of 2022) sets the long-term mitigation targets. Brazil had no binding carbon price during the sample: the Brazilian Emissions Trading System (SBCE) was created by Law 15,042 of December 2024 and is in its implementation phase. These features determine the carbon-price

component of the climate state: it is constant at zero for Brazil, nearly constant within years for the other markets, and changes mainly through discrete policy decisions.

2.3 Distributionally Robust Optimization (DRO) and State-Space Models in Portfolios

The foundations of DRO with Wasserstein ambiguity sets were established by Mohajerin Esfahani and Kuhn (2018), Blanchet and Murthy (2019) and Gao and Kleywegt (2023). Kuhn et al. (2019) and Rahimian and Mehrotra (2022) survey the field. In portfolio selection, Pflug et al. (2012) show that the equally weighted portfolio is optimal under high model ambiguity, Blanchet et al. (2022) relate Wasserstein robustness in mean-variance (MV) selection to norm regularization, and Lotfi and Zenios (2018) study robust CVaR under joint ambiguity in distributions, means and covariances. These results imply that the geometry of the transport cost and the size of the radius determine whether a robust portfolio differs from simple diversification rules, which are difficult to beat out of sample (DeMiguel et al., 2009). Conditioning the nominal distribution on covariates is the idea behind prescriptive analytics (Bertsimas and Kallus, 2020), whose kernel weights are Nadaraya–Watson estimators of the conditional distribution (Nadaraya, 1964; Watson, 1964). The finite-sample guarantees of Wasserstein DRO depend on the radius (Gao, 2023), which in practice is selected by validation (Mohajerin Esfahani and Kuhn, 2018).

Climate variables are observed with noise, with missing values (carbon prices) and at mixed frequencies. A linear Gaussian state-space model handles these features, and its parameters can be estimated by expectation-maximization with the Rauch-Tung-Striebel smoother (Shumway and Stoffer, 1982; Durbin and Koopman, 2012). Because the model is time-invariant, the posterior covariance of the filter converges to the stationary solution of the Riccati equation regardless of the data (Durbin and Koopman, 2012); the data-dependent information is in the innovations, which we treat as climate news, in line with the use of climate news in asset pricing (Engle et al., 2020).

3. Methodological Framework

For each market $c \in \{\text{BR, MX, CO, CL}\}$, an investor chooses at the end of month t a weight vector $w_t \in \mathcal{W}_t \subset \Delta^N$ (no short sales) for month $t+1$. Monthly simple returns in USD are $r_{t+1} \in \mathrm{R}^N$ and the portfolio loss is $\ell_w(r) = -w^\top r$. The risk criterion is the conditional value at risk at level α

$$CVaR_\alpha^{\mathrm{Q}}(\ell_w) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{\alpha}\, \mathbb{E}_{\mathrm{Q}}[(\ell_w(r) - \tau)^+] \right\}. \tag{1}$$

Because the distribution $\mathbb{Q}$ of next month's returns is unknown, the robust decision minimizes the worst case over an ambiguity ball centered at a nominal measure $\hat{P}_t$:

$$w_t^\star = \arg\min_{w \in \mathcal{W}_t} \sup_{Q:\, W_2(Q, \hat{P}_t) \le \varepsilon_t} CVaR_\alpha^{\mathrm{Q}}(\ell_w), \tag{2}$$

$$W_2(Q,P) = \left(\inf_{\gamma \in \Gamma(Q,P)} E_\gamma \left[(r - r')^\top M_t (r - r') \right] \right)^{1/2}.$$

Climate enters three objects: the nominal measure $\hat{P}_t$, the transport metric M_t and the feasible set $\mathcal{W}_t$. The economic reading is direct: the Wasserstein ball contains the distributions that the investor cannot rule out with the available evidence; climate determines which historical scenarios are relevant today, along which directions misspecification is more plausible, and which portfolios are admissible given their climate exposure.

3.1 Climate State Vector

For each market, the observed climate state is

$$\mathbf{y}_t^{(c)} = (P_C^{(c)}(t),\ T_{\text{anom}}^{(c)}(t),\ MEI(t),\ log\text{Comm}^{(c)}(t))^\top. \tag{3}$$

where $P_C^{(c)}$ is the explicit carbon price in USD/tCO$_2$e, $T_{\text{anom}}^{(c)}$ is the NASA GISTEMP v4 temperature anomaly averaged over the country's central economic region, MEI is NOAA's Multivariate ENSO Index v2 and Comm is the price of Brent (Brazil, Colombia), WTI (Mexico) or copper (Chile). The variables are used in their natural, unbounded scales: asinh(P_C) admits zero prices and behaves as log($2P_C$) for high prices. Climate variables are lagged one month to reflect publication delays. A component without variation in the estimation sample (Brazil's carbon price) is excluded from the state, because it carries no information and makes the transition covariance singular.

The nominal measure (Section 3.3) is conditioned on stationary features f_t rather than on levels: the MEI level (ENSO phase), its three-month change (direction of the phase), the detrended temperature anomaly (trend estimated with data up to $t-1$) and the three-month log return of the commodity. If a variable has a trend, $x_s = a + bs + u_s$, the distance $|x_s - x_t| \approx |b|\,|t-s|$ measures proximity in time rather than similarity of states, so a kernel on trending levels amounts to weighting by recency

3.2 State-Space Model and Climate News

Let $y_t \in \mathbb{R}^k$ be the standardized climate state. The model is

$$x_t = a + \Phi x_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q); \qquad y_t = x_t + v_t, \quad v_t \sim \mathcal{N}(0, R), \tag{4}$$

with R diagonal. The Kalman filter yields the prediction $x_{t|t-1}$, the innovation $\nu_t = y_t - x_{t|t-1}$ with covariance $S_t = P_{t|t-1} + R$, the gain $K_t = P_{t|t-1} S_t^{-1}$ and the update $x_{t|t} = x_{t|t-1} + K_t \nu_t$; with missing observations the update uses only the observed rows. The parameters (a, Φ, Q, R) are estimated by EM (Shumway and Stoffer, 1982) on an expanding window re-estimated at the end of each year, and the spectral radius of Φ is bounded at 0.97. Because the system is time-invariant, $P_{t|t}$ converges to its stationary Riccati value independently of the data, so the posterior covariance is not informative about the climate state. The innovations νt are the climate news of the month, the part of the observation not

anticipated by the model, and they are the channel through which climate enters the transport metric.

3.3 Climate-Conditioned Nominal Measure

The scenario library contains all complete months since January 2014 (expanding). Each historical scenario s receives the weight

$$\pi_s(t) = \frac{1-\omega}{n} + \omega \frac{K(d_s/h)}{\sum_u K(d_u/h)}, \qquad d_s^2 = (f_{s-2} - f_{t-1})^\top S_f^{-1} (f_{s-2} - f_{t-1}), \tag{5}$$

with Gaussian kernel $K(u) = e^{-u^2/2}$, S_f the covariance of the features, bandwidth $h = 1$ and $\omega = 0.5$. Timing is symmetric: the return of month s, decided at the end of $s-1$, is compared through the features known at $s-1$ (published with one month of delay), just as the decision at t uses ft−1. The mixture with the uniform measure keeps the effective sample size, $1/\sum_s \pi_s^2$, away from degenerate values. Economically, if the Pacific is today in a warm phase that is intensifying, the historical months with that phase receive more weight in the evaluation of the loss tail; with $\omega = 0$ the uniform empirical measure is recovered.

3.4 Transport Metric with a Climate Block

The transport cost is $c_t(r, r') = (r - r')^\top M_t (r - r')$ with

$$M_t^{-1} = \Sigma_t^{\mathrm{OAS}} + \kappa\, B_t \Lambda_t B_t^\top, \tag{6}$$

where Σ_t^{OAS} is the oracle-approximating shrinkage covariance of the library (Chen et al., 2010), $B_t (N \times k)$are ridge betas of returns on the standardized climate innovations, Λt is the exponentially weighted covariance of the innovations (half-life of six months) and $\kappa = 1$. Both terms are in squared return units. Because moving a scenario by δ costs $\delta^\top M_t \delta$, directions with larger climate-induced variance are cheaper for the adversary: the ball is elongated along the directions in which climate news moves returns jointly. With $\kappa = 0$ the metric is the OAS Mahalanobis metric.

3.5 Exact Worst Case and Conic Reformulation

The objective is to choose the weights **w** that minimize CVaR under the worst-case scenario contained in the type-2 Wasserstein ball.

Proposition 1 (Worst-case CVaR over a type-2 Wasserstein ball). Let $M \succ 0$ with $M^{-1} = GG^\top$, $\hat{P} = \sum_s \pi_s \delta_{r_s}$ with support in $\mathbb{R}^N$, and $\ell_w(r) = -w^\top r$. Then, for every $\varepsilon \geq 0$,

$$\sup_{Q:\, W_2(Q,\hat{P}) \leq \varepsilon} CVaR_\alpha^Q(\ell_w) = CVaR_\alpha^{\hat{P}}(\ell_w) + \frac{\varepsilon}{\sqrt{\alpha}} \|G^\top w\|_2. \tag{7}$$

Proof. Upper bound. Use the dual representation $CVaR_\alpha(\ell) = \sup\{\mathbb{E}[\zeta \ell] : 0 \leq \zeta \leq 1/\alpha,\ \mathbb{E}\zeta = 1\}$. For Q in the ball there is a coupling γ of (r, r'), $r \sim \hat{P}$, $r' \sim Q$, with

$\mathbb{E}_\gamma\|r'-r\|_M^2 \le \varepsilon^2$. For any admissible $\zeta(r')$, $\mathbb{E}_\gamma[\zeta\,\ell_w(r)] \le CVaR_\alpha^{\hat{P}}(\ell_w)$ because $\mathbb{E}_\gamma[\zeta|r]$ is also admissible, and by the Cauchy-Schwarz inequality $\ell_w(r') - \ell_w(r) = -w^\top(r'-r) \le \|G^\top w\|_2\,\|r'-r\|_M$. Since $\mathbb{E}\zeta^2 \le \mathbb{E}\zeta/\alpha = 1/\alpha$, $\mathbb{E}_\gamma\big[\zeta\big(\ell_w(r') - \ell_w(r)\big)\big] \le \|G^\top w\|_2\,\varepsilon/\sqrt{\alpha}$. Attainment. Moving the α-mass of the loss tail of $\hat{P}$ by $\delta = -c\,M^{-1}w/\|G^\top w\|_2$ with $c = \varepsilon/\sqrt{\alpha}$ costs $\alpha c^2 = \varepsilon^2$ and raises every tail loss by $c\,\|G^\top w\|_2$. The value coincides with the Lagrangian dual of the Wasserstein problem (Blanchet and Murthy, 2019; Gao and Kleywegt, 2023; Kuhn et al., 2019).

The robust problem is therefore the second-order cone program (Lobo et al., 1998)

$$\min_{w\in\mathcal{W}_t,\,\tau,\,u\ge 0}\quad \tau + \frac{1}{\alpha}\sum_s \pi_s(t)\,u_s + \frac{\varepsilon}{\sqrt{\alpha}}\,\|G_t^\top w\|_2 \quad \text{s.t.} \quad u_s \ge -r_s^\top w - \tau. \tag{8}$$

The worst case is the nominal CVaR plus a penalty proportional to the portfolio volatility measured with M_t^{-1}, and ε is the root-mean-square displacement granted to the adversary in Mahalanobis units, hence dimensionless. As $\varepsilon \to 0$ the decision minimizes the π-weighted CVaR and climate acts through the scenarios that form the tail; as $\varepsilon \to \infty$ it converges to the minimum-variance portfolio under $\Sigma_t^{\mathrm{OAS}} + \kappa B\Lambda B^\top$ and climate acts only through the metric. With M = I the penalty is $\varepsilon\alpha^{-1/2}\,\|\,w\,\|\,2$, which is minimized by equal weights on the simplex; this is the mechanism by which Euclidean Wasserstein DRO converges to $1/N$ (Pflug et al., 2012).

Remark 1 (Nesting). DRO-CC with $\omega = 0$ and $\kappa = 0$ coincides with a covariance-based (OAS) Wasserstein DRO, and any of these models with $\varepsilon = 0$ coincides with the empirical CVaR portfolio. The comparison between DRO-CC and its climate-free counterpart therefore isolates the contribution of climate information.

At each rebalancing date, ε is selected from a grid of twelve values, $\varepsilon \in \{0, 0.01, 0.025, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 6.4, 12.8\}$, by rolling-origin cross-validation: for each value, the decisions the model would have taken at each of the 24 previous month-ends are rebuilt with the information of each date, and the CVaR at 20% of the 24 realized one-month returns is computed; the smallest criterion is chosen (ties go to the smaller radius). A 20% criterion over 24 months rests on about five tail observations, which reduces the variance of the selection. We also report the realized out-of-sample CVaR as a function of a fixed radius.

3.6 Firm-Level Climate Exposures

Each issuer i has a transition exposure CI_i and a physical exposure $PHYS_i$. Audited Scope 1+2 intensities are not openly available for the full universe before the evaluation period. CI_i is therefore an expert-based prior on [0,1] (see Table 3), which we state as such. $PHYS_i$ is measured from data. Prices respond to news about cash flows, not to the level of a known variable: if the MEI follows an autoregressive process with persistence ϕ, the correlation between returns that respond to $\Delta\mathrm{MEI_t}$ and the MEI level is attenuated by $corr(\mathrm{MEI_t}, \Delta\mathrm{MEI_t}) = p(1-\phi)/2$. For each issuer we estimate

$$r_{it} = a_i + \sum_{l=0}^{3} g_{il}\, \Delta \mathrm{MEI}_{t-l} + c_i\, m_{-i,t} + e_{it}, \tag{9}$$

where $m_{-i,t}$ is the equally weighted return of the rest of the market, which isolates the withinmarket (differential) exposure relevant for allocation and absorbs common shocks. With $R^2_{p,i}$ the partial R^2 of the ENSO block and $\mathrm{F_i}$ its F statistic, the ENSO component is shrunk continuously

$$\phi_i^{\mathrm{ENSO}} = \sqrt{R^2_{p,i}\,(1 - 1/F_i)_+}, \quad \mathrm{PHYS}_i = 0.5\phi_i^{\mathrm{ENSO}} + 0.5\phi_i^{\mathrm{water}}, \tag{10}$$

so that it vanishes when the evidence is weak ($F_i \leq 1$). ϕ_i^{water} averages the WRI Aqueduct 4.0 baseline water stress, interannual variability and drought-risk scores (0-5 scale divided by 5) over the first-level regions where the firm operates, weighted by area (Kuzma et al., 2023). Financial, telecommunication and holding firms receive their headquarters country. PHYS is first estimated with data for 2014–2017 and re-estimated at the end of each year with an expanding window.

3.7 Feasible Set

The feasible set combines transition and physical constraints:

$$\mathcal{W}_t = \{w \geq 0:\ w^{\top}\mathbf{1} = 1,\ w \leq w_{\max}, \mathrm{CI}^{\top}w \leq (1-\rho)(1-g)^{\tau_t}\,\mathrm{CI}^{\top}b, \\ \mathrm{PHYS}^{\top}w \leq (1 - \eta\,\mathbb{1}\{|\mathrm{MEI}_{t-1}| > 1\})\,\mathrm{PHYS}^{\top}b\} \tag{11}$$

with b the equally weighted benchmark, $\rho = 0.5$ and $g = 0.07$ (initial reduction and annual self-decarbonization of an EU Paris-aligned benchmark, European Commission, 2020), τ_t the years since the start of the evaluation, $w_{max} = 0.35$ and $\eta = 0.10$. Because there is no regulatory standard for physical exposure, the portfolio's physical exposure may not exceed the benchmarks, and it is tightened by 10% in strong ENSO episodes. If the physical limit is below its attainable minimum given the other constraints, it is set at that minimum plus 10% of the margin to the benchmark. If the set is still empty, the cap and then the physical constraint is relaxed, and every relaxation is recorded. Rescaling exposures by a common factor on both sides of a constraint leaves it unchanged, so no dynamic rescaling is applied.

Algorithm 1 summarizes the computational architecture.

Algorithm 1. DRO-CC portfolio optimization with a climate set

Require Monthly returns; climate series $(P_C, T_{\mathrm{anom}}, \mathrm{MEI}, \mathrm{Comm})$ lagged one month; CI prior; Aqueduct scores; parameters (Table 10).

1: **for** each market c and each quarterly rebalancing date t from Dec-2017 to Sept-2025 **do**

2: **if** t is a year end

3: **then** re-estimate the state-space model by EM and PHYS by Eq. (9)-(10) with data up to t

4: **endif**

5: Filter the climate state; compute innovations ν_s, betas B_t and EWMA covariance Λ_t

6: Build $\pi(t)$ (Eq. 5) and M_t^{-1} (Eq. 6) on the expanding library

7: Compute the limits of $\mathcal{W}_t$ (Eq. 11)

8: Select ε_t by rolling-origin cross-validation over the previous 24 months

9: $w_t \leftarrow$ solve the SOCP (Eq. 8); hold weights (with price drift) until the next rebalancing

10: **end for**

11: Compute net-of-cost metrics (per unit of turnover) and bootstrap/DM inference against the benchmarks.

4. Empirical Application and Analysis of Results

4.1 Data, Asset Universe, and Backtesting Protocol

This section implements the proposed approach. The universe comprises $N_{total} = 55$ highly liquid stocks, distributed across four markets: $N = 15$ in Brazil (B3), $N = 14$ in Mexico (BMV), $N = 13$ in Colombia (BVC), and $N = 13$ in Chile (BCS). Table 1 details the selected stocks. The suffixes identify the listing exchange: .SA (B3, Sao Paulo), .MX (BMV, Mexico), .CL (BVC, Colombia), and .SN (Santiago Exchange, Chile). The universe was selected with market capitalization above USD 500 million in 2020; because this criterion uses information after the start of the evaluation, it may favor surviving firms. The bias is common to all strategies and to equally weighted benchmarks, so it affects levels more than comparisons, but it is a limitation of the design.

Table 1. Issuers, tickers, and sectors by market

Ticker	Issuer	Sector
Brazil — IVOBESPA / B3 ($N = 15$)		
PETR4.SA	Petrobras	Oil and gas
VALE3.SA	Vale	Mining (iron, nickel)
ITUB4.SA	Itaú Unibanco	Banking
BBDC4.SA	Bradesco	Banking
CPFE3.SA	CPFL Energia	Power distribution and generation
CMIG4.SA	Cemig	Hydropower generation
CSNA3.SA	CSN	Integrated steelmaking
WEGE3.SA	WEG	Capital goods / motors
SUZB3.SA	Suzano	Pulp and paper
GGBR4.SA	Gerdau	Steel
EGIE3.SA	Engie Brasil	Hydropower generation
ABEV3.SA	Ambev	Beverages
BBAS3.SA	Banco do Brasil	Banking

Ticker	Issuer	Sector
TOTS3.SA	TOTVS	Technology (software)
RENT3.SA	Localiza	Car rental
Mexico — IPC / BMV ($N = 14$)		
FEMSAUBD.MX	FEMSA	Beverages and retail
GMEXICOB.MX	Grupo México	Copper mining
WALMEX.MX	Walmart de México	Retail
GAPB.MX	Grupo Aeroportuario del Pacífico	Airport infrastructure
AMXB.MX	América Móvil	Telecommunications
RA.MX	Regional (Banregio)	Banking
GRUMAB.MX	Gruma	Food (corn flour)
KIMBERA.MX	Kimberly-Clark México	Tissue paper
CEMEXCPO.MX	CEMEX	Cement
GFNORTEO.MX	Grupo Financiero Banorte	Banking
BIMBOA.MX	Grupo Bimbo	Food (baking)
ORBIA.MX	Orbia	Chemicals and plastics
AC.MX	Arca Continental	Beverages
TLEVISACPO.MX	Grupo Televisa	Media/Telecommunications
Colombia — COLCAP / BVC ($N = 13$)		
ECOPETROL.CL	Ecopetrol	Oil and gas
PFCIBEST.CL	Bancolombia (pref.)	Banking
PFGRUPOARG.CL	Grupo Argos (pref.)	Cement and energy
NUTRESA.CL	Grupo Nutresa	Processed foods
CELSIA.CL	Celsia	Power generation
GRUPOSURA.CL	Grupo Sura	Financial services
TERPEL.CL	Terpel	Fuel distribution
GEB.CL	Grupo Energía Bogotá	Energy and gas
CEMARGOS.CL	Cementos Argos	Cement
PFDAVVNDA.CL	Davivienda (pref.)	Banking
ISA.CL	ISA	Power transmission
EXITO.CL	Grupo Éxito	Retail
PFAVAL.CL	Grupo Aval (pref.)	Banking
Chile — IPSA / Santiago Exchange ($N = 13$)		
SQM-B.SN	Sociedad Química y Minera (SQM)	Lithium and mining
BSANTANDER.SN	Banco Santander Chile	Banking
COPEC.SN	Empresas Copec	Energy and fuels
FALABELLA.SN	Falabella	Retail
CENCOSUD.SN	Cencosud	Retail
CMPC.SN	Empresas CMPC	Pulp and paper
COLBUN.SN	Colbún	Hydropower generation
CCU.SN	CCU	Beverages

Ticker	Issuer	Sector
LTM.SN	LATAM Airlines	Air transport
ENTEL.SN	Entel	Telecommunications
QUINENCO.SN	Quiñenco	Conglomerate (holding)
ENELAM.SN	Enel Américas	Power generation
ECL.SN	Engie Energía Chile	Thermal and renewable

Source: own elaboration with data from B3, BMV, BVC and Bolsa de Santiago.

All series come from public sources (Table 2). Prices are month-end adjusted closes converted to USD. The risk-free rate is the 13-week U.S. Treasury bill. Carbon prices for Mexico, Colombia and Chile come from the World Bank Carbon Pricing Dashboard and the official tax resolutions. Brazil's explicit carbon price is zero throughout the sample.

Table 2. Data sources and periods

Variable	Source/ URL	Period
Stock prices, FX, T-bill	Yahoo Finance, adjusted month-end closes	2014–2025
Commodities	Yahoo Finance (BZ=F, CL=F, HG=F)	2014–2025
Country ETFs	iShares EWZ, EWW, ECH, ICOLCAP	2018–2025
ENSO MEI v2	NOAA PSL, Multivariate ENSO Index v2	2010–2025
T_{anom}	NASA GISTEMP v4(GHCN v4 + ERSST v5), 1200 km, via NOAA PSL	2010–2025
CI^{1+2}	Expert base-prior (based on CDP, and IPCC AR6 WG3)	2010–2025
ϕ^{water}	WRI Aqueduct 4.0 (baseline annual)	2014–2025
ϕ^{ENSO}	Eq. 9 and MEI v2	2014–2025
Carbon Prices	World Bank Carbon Pricing Dashboard; DIAN Resolutions 000007 of 2021 and 000008 of 2025 (Colombia); Law 20,780 (Chile); Law 15,042 of 2024 (Brazil, no price)	2010–2025

Source: own elaboration.

4.2.2 Estimation and backtesting protocol.

The climate-state starts in January 2010 and stock returns in January 2014. The first decision is taken at the end of December 2017 (the in-sample – IS period), so the evaluation (the out-of-sample – OOS period) covers the 96 months from January 2018 to December 2025, which include the 2018–2019 weak El Nino, the 2020–2023 La Nina sequence, the 2023–2024 El Nino, the pandemic crash and the 2022 monetary tightening. Portfolios are rebalanced quarterly, transaction costs are 25 bp per unit of turnover, and all strategies share the same scenario, library and feasible set.

Seven strategies are compared: $1/N - \text{restr}$ (Euclidean projection of equal weights onto $\mathcal{W}_t$); MVO-OAS (minimum variance with OAS covariance). CVaR ($\varepsilon = 0$); DRO-Eucl (CV) (Euclidean metric, uniform measure); DRO-OAS (CV) (OAS metric, uniform measure). The proposed DRO-CC; and the DRO-KF (Euclidean penalty with the fixed in-sample radius).

The country ETF (for each stock market index) is reported as an external benchmark. As a robustness design, the model is also estimated with data up to 2021 and evaluated over 2022–2025 with PHYS fixed at its 2021 value.

4.2 Firm-Level Climate Exposures

Tables 3a and 3b report the parameters of the main specification, as well as the following for each issuer, the transition prior CI, the physical exposure assigned in the submitted version and the physical exposure measured from data at the first decision (Dec-2017), with its ENSO component.

Table 3a. Parameters of the main specification

Parameter	Value	Definition
Climate / returns	2010-01 / 2014-01	First month of the climate state / first stock return
First decision; evaluation	2017-12; 2018-01 to 2025-12	96 out-of-sample months
Publication lag	1 month	Lag of climate variables
Rebalancing	Quarterly; 25 bp	Cost per unit of turnover
Scenario library	Expanding from 2014	Minimum of 36 complete months per asset
State-space	Annual, expanding	EM, spectral radius bounded at 0.97
CVaR level	0.05	Tail level of the objective
Radius grid	0 to 12.8 (12 values)	Mahalanobis units
Cross-validation	24 months, CVaR at 20%	Minimum of 12 months
Kernel	$\omega = 0.5$, $h = 1$	Features: MEI, Δ3MEI, detrended T, 3-month commodity return
Climate block	$\kappa = 1$; EWMA half-life 6; ridge 1	Eq. (6)
Transition constraint	$\rho = 0.5$, $g = 0.07$	EU Paris-aligned benchmark
Physical constraint	$\leq$ benchmark; $\eta = 0.10$	Slack of 10% of the margin if bounded
Position cap	0.35	Maximum weight per asset
PHYS	3 lags, market control, weight 0.5	Re-estimated annually, expanding
Bootstrap	2.000 replications	Automatic block; sensitivity 3, 6, 12

Source: own elaboration.

Table 3b. Firm-level climate exposures

Ticker	CI	PHYS	ϕ^{ENSO}
Brazil (B3)			
PETR4.SA	0.92	0.200	0.110
VALE3.SA	0.78	0.122	0.000

Ticker	CI	PHYS	ϕ^{ENSO}
ITUB4.SA	0.08	0.119	0.000
BBDC4.SA	0.08	0.119	0.000
CPFE3.SA	0.12	0.336	0.346
CMIG4.SA	0.12	0.217	0.131
CSNA3.SA	0.88	0.142	0.000
WEGE3.SA	0.38	0.222	0.129
SUZB3.SA	0.42	0.162	0.000
GGBR4.SA	0.85	0.282	0.251
EGIE3.SA	0.10	0.279	0.243
ABEV3.SA	0.35	0.119	0.000
BBAS3.SA	0.07	0.119	0.000
TOTS3.SA	0.05	0.217	0.196
RENT3.SA	0.15	0.168	0.098
Mexico (BMV)			
FEMSAUBD.MX	0.32	0.278	0.000
GMEXICOB.MX	0.80	0.458	0.289
WALMEX.MX	0.10	0.278	0.000
GAPB.MX	0.10	0.508	0.336
AMXB.MX	0.18	0.278	0.000
RA.MX	0.05	0.278	0.000
GRUMAB.MX	0.40	0.278	0.000
KIMBERA.MX	0.35	0.399	0.244
CEMEXCPO.MX	0.96	0.453	0.351
GFNORTEO.MX	0.05	0.278	0.000
BIMBOA.MX	0.28	0.343	0.131
ORBIA.MX	0.72	0.278	0.000
AC.MX	0.30	0.480	0.404
TLEVISACPO.MX	0.10	0.384	0.212
Colombia (BVC)			
ECOPETROL.CL	0.95	0.228	0.293
PFCIBEST.CL	0.06	0.071	0.000
PFGRUPOARG.CL	0.82	0.243	0.343
NUTRESA.CL	0.25	0.164	0.186
CELSIA.CL	0.15	0.175	0.135
GRUPOSURA.CL	0.05	0.155	0.167
TERPEL.CL	0.45	0.071	0.000
GEB.CL	0.10	0.144	0.146
CEMARGOS.CL	0.90	0.157	0.172
PFDAVVNDA.CL	0.06	0.071	0.000
ISA.CL	0.08	0.071	0.000
EXITO.CL	0.08	0.132	0.121
PFAVAL.CL	0.05	0.071	0.000
Chile (BCS)			

Ticker	CI	PHYS	ϕ^{ENSO}
BSANTANDER.SN	0.05	0.341	0.000
COPEC.SN	0.65	0.225	0.000
FALABELLA.SN	0.08	0.223	0.000
CENCOSUD.SN	0.08	0.225	0.000
CMPC.SN	0.40	0.225	0.000
COLBUN.SN	0.05	0.202	0.000
CCU.SN	0.30	0.403	0.195
LTM.SN	0.72	0.331	0.212
ENTEL.SN	0.10	0.470	0.489
QUINENCO.SN	0.22	0.225	0.000
ENELAM.SN	0.18	0.383	0.316
ECL.SN	0.45	0.167	0.000

CI: expert-based transition prior. PHYS: measured exposure, Eq. (10). ϕ^{ENSO}: ENSO component of PHYS. Source: own elaboration with data from Yahoo Finance, NOAA PSL (MEI v2) and WRI Aqueduct 4.0.

Table 4. Climate exposures by market

	CI	PHYS			ϕ^{ENSO}	ϕ^{water}
Market (N)	**Mean**	**Mean**	**St. Dev**	**Max PHYS**	**Mean**	**Mean**
Brazil ($N = 15$)	0.357	0.429	0.070	CPFL Energia	0.100	0.276
Mexico ($N = 14$)	0.336	0.311	0.089	GAP	0.141	0.569
Colombia ($N = 13$)	0.308	0.341	0.061	Grupo Argos	0.120	0.149
Chile ($N = 13$)	0.289	0.440	0.124	Engie Chile	0.131	0.485

Note: Market means cross-sectional standard deviation of PHYS, issuer with the highest PHYS and number of issuers with a significant ENSO news block at 5%.
Source: own elaboration based on Table 3.

Table 4 summarizes the four markets and Figure 1 shows the position of the 55 issuers in the transition–physical plane. Three features stand out. First, measured physical exposure is lower and much less dispersed than the assigned values: the cross-sectional standard deviation of PHYS within markets is 0.06-0.12, against 0.24-0.36 for the assigned values. Second, the ranking of markets changes. Mexico and Chile have the highest measured exposure, driven by water stress in their regions of operation ϕ^{water} of 0.57 and 0.49), whereas Brazil and Colombia, with lower water stress, move to the bottom; the submitted version ranked Chile and Brazil first. The rank correlation between the prior and the measured values is 0.27 ($p = 0.05$). Third, the ENSO component is small and rarely significant. Hydropower generators, which the prior placed at the top (Colbun 0.96, Engie Brasil 0.92), do not show a differential response to ENSO news once the common market component is controlled for.

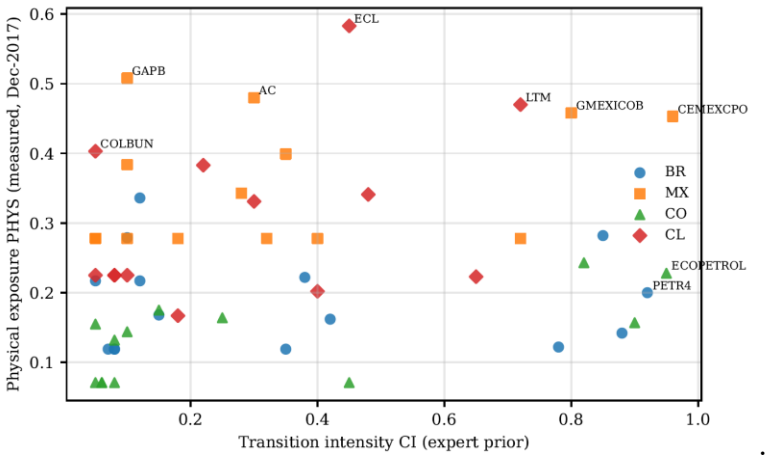


Figure 1. Transition prior (CI) and measured physical exposure (PHYS, December 2017) of the 55 issuers. Source: own elaboration.

4.3 ENSO Series from NOAA's MEI v2

The third component of the state vector, ENSO(t), is built from the Multivariate ENSO Index version 2 (MEI v2) of NOAA, which summarizes in a single index five ocean-atmosphere variables of the tropical Pacific (sea-level pressure, the zonal and meridional components of the surface wind, sea-surface temperature, and outgoing longwave radiation). The MEI v2 is reported as overlapping bimonthly means, which are reindexed to monthly frequency over the study's common calendar and completed, where missing, by linear interpolation and forward fill, preserving the directionality of the index (positive for El Niño, negative for La Niña). Table 5 reports the number of issuers for which the null of no response is rejected at 5% under several specifications, together with the probability of observing at least that many rejections under the null.

Table 5. Rejections of the null of no ENSO response (55 issuers)

	2014–2017		2014–2025	
Specification	**Rejections**	**P(X ≥ k)**	**Rejections**	**P(X ≥ k)**
Contemporaneous MEI level, no market control	4	0.295	3	0.523
Contemporaneous MEI level, market control	3	0.523	2	0.768
ENSO news, lags 0–3, no market control	9	0.002	1	0.940
ENSO news, lags 0–3, market control (main)	4	0.295	1	0.940
ENSO news, lags 0–3, market control, local currency	5	0.140	1	0.940
ENSO news, lags 0–6, market control	7	0.019	4	0.295
Regime |MEI| > 1 interaction, market control	2	0.768	4	0.295

Note: P(X ≥ k) is binomial with 55 independent trials and success probability 0.05; with cross-sectional dependence it overstates the significance of the count.
Source: own elaboration with data from NOAA PSL.

The table supports three conclusions. (a) Measuring the response to ENSO news rather than to the MEI level is appropriate: without market control, the news specification detects 9

responses in 2014-2017 against 4 for the level, consistent with the attenuation factor $\sqrt{(1-\phi)/2} = 0.16$ implied by the MEI persistence (ϕ = 0.949; the observed correlation between the MEI level and its change is 0.146). (b) The response is mostly common to the market: controlling for the rest-of-market return reduces the rejections from 9 to 4, a count compatible with chance. PHYS measures the within-market exposure that matters for allocation, and this exposure is close to zero. (c) The response is episodic: over 2014-2025 it disappears (one rejection), which suggests that the extreme warm event of 2015-2016 dominates the initial window; in the annual re-estimations from 2017 to 2024 the number of significant issuers (4, 2, 3, 2, 1, 2, 2, 1) never exceeds what chance would produce. The ENSO threshold that tightened constraints in the submitted version (|MEI|/3 > 0.8, i.e. |MEI| > 2.4) is never reached in 2014–2025: the largest |MEI| in 2023–2025 is 1.22, which explains why the ENSO channel was inactive in that version.

4.4 Model Specification and Hypotheses

The hypotheses were stated before the evaluation. Let $\Delta CVaR^{A,B} = CVaR_{5\%}(r^A) - CVaR_{5\%}(r^B)$ be the difference in monthly CVaR at 5% between strategies A and B (negative values favor A), and $d_t^{A,B} = L(r_t^A) - L(r_t^B)$ the difference in the continuous tail loss $L(r) = (\min\{r, 0\})^2$.

- **H1**: (value of climate information). $\Delta CVaR^{\text{DRO-CC,DRO-OAS}} < 0$ and $\mathbb{E}\left[d_t^{\text{DRO-CC,DRO-OAS}}\right] < 0$.
- **H2**: (value of the climate-conditioned robust model): DRO-CC has lower tail risk than MVO-OAS and than $1/N$-restr.
- **H3**: (physical-exposure ordering): the gain in H1 is larger in the markets with the highest measured physical exposure at the first decision (Mexico and Chile, Table 4).

Each comparison uses stationary bootstrap confidence intervals (Politis and Romano, 1994) for ΔCVaR, ΔSharpe and ΔMaxDD, with the mean block length chosen by the automatic rule of Politis and White (2004) as corrected by Patton et al. (2009) and sensitivity to fixed blocks of 3, 6 and 12 months. The Diebold–Mariano test (Diebold and Mariano, 1995) on dt with the Harvey et al. (1997) correction and Newey-West variance (Newey and West, 1987); and Holm adjustment across markets (Holm, 1979). A pooled panel resamples dates jointly across the four markets. A narrow interval around zero is informative (the strategies do not differ by more than a given margin), a wide interval indicates lack of power.

4.5 Feasible Set, Radius and Filter Diagnostics

Table 6 reports how the constraints, the radius and the climate block operate in DRO-CC. All 15.422 optimization problems were solved to optimality, the closed form of Proposition 1 matches the Lagrangian dual to within 5×10^{-9} at the first decision of each market, and the feasible set never had to be relaxed. With the EU Paris-aligned threshold the most diversified feasible portfolio at the first decision has 9.1-11.5 effective positions, against 5.2-6.6 under

the 70% reduction of the submitted version, so the constraints no longer force concentration. The transition constraint binds at every rebalancing for all strategies, and the physical constraint binds for DRO-CC at 100% of the dates in Brazil, 66% in Colombia and about 30% in Mexico and Chile; the measurement of PHYS therefore shapes the portfolios.

Table 6. Constraints, radius and climate block in DRO-CC (2018–2025)

	Brazil	México	Colombia	Chile
Max. N_{eff} attainable at first decision	11.50	10.21	10.76	9.07
Transition constraint active (% of rebal.)	100	100	100	100
Physical constraint active (%)	100.0	28.1	65.6	31.2
Position cap active (%)	71.9	87.5	84.4	75.0
Median selected ε	1.0	0.8	9.6	0.4
at the upper end of the grid (%)	43.8	9.4	50.0	40.6
Penalty share of the objective (mean)	0.57	0.56	0.81	0.60
Climate share of $tr(M^{-1})$	0.13 (0.42)	0.11 (0.49)	0.16 (0.53)	0.10 (0.20)

Note: Penalty share: $\varepsilon\alpha^{-1/2} \parallel G^{\top}w \parallel 2$ over the objective of Eq. (8) at the selected radius. Climate share: $\kappa tr(B\Lambda B^{\top})/tr(M^{-1})$.
Source: own elaboration.

The radius is weakly identified. Cross-validation selects the upper end of the grid at 41–50% of the dates in Brazil, Colombia and Chile, and its criterion is nearly flat (the best and worst radius differ by a median 6–9% of the criterion). Figure 2 shows that the radius that minimizes out-of-sample CVaR differs across markets: small in Brazil ($\varepsilon \leq 0.1$, CVaR 17.1%) and Chile (0.2; 16.3%), intermediate in Mexico (0.8; 16.8%) and at the top of the grid in Colombia (12.8; 17.0%). Cross-validation matches the best fixed radius in Mexico and Colombia, but in Brazil and Chile it leads to the minimum-variance regime, where the realized CVaR is 0.7 and 0.3 points worse. With penalty shares of 0.56–0.81, DRO-CC operates most of the time near the minimum-variance portfolio under $\Sigma_t^{\mathrm{OAS}} + \kappa B\Lambda B^{\top}$, where the kernel loses influence. Choosing a fixed radius from Figure 2 would fit the design to the evaluation period, so the cross-validated rule is kept as the main specification.

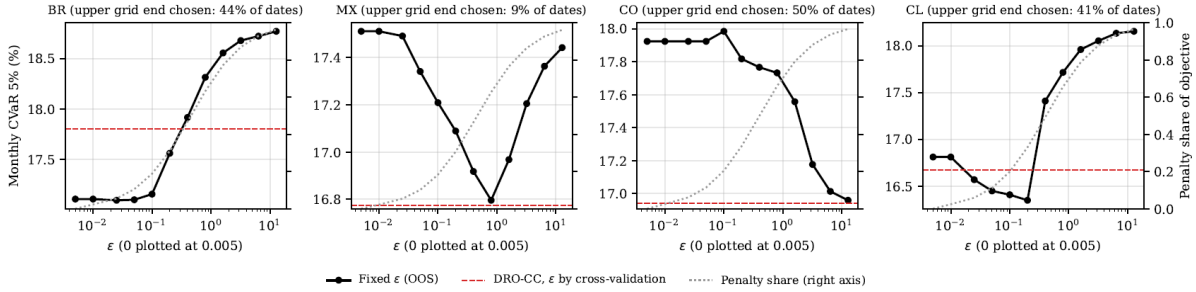


Figure 2. Out-of-sample monthly CVaR at 5% of DRO-CC with a fixed radius (solid line) and with the cross-validated radius (dashed line) and share of the Wasserstein penalty in the objective (dotted line, right axis).

Source: own elaboration.

The climate block is not negligible in the metric: it accounts on average for 10-16% of $tr(M^{-1+t})$, and for up to half of it at specific dates. The kernel re-weights the library moderately: the effective sample size is 61-66 scenarios out of a library of 87-95 months.

The stationary features remove the recency bias that a kernel on levels would produce (rank correlation between weights and recency at the first decision of 0.15, 0.24, 0.40 and −0.13, against 0.46, 0.70, 0.44 and 0.05 with levels). Three numerical aspects of the state-space model are reported for transparency: the EM algorithm reaches its 200-iteration limit in 35 of 40 estimations; the stability bound $\rho(\Phi) = 0.97$ binds in 20 of 40, because detrended temperature and commodity prices are nearly integrated; and Mexico's transition covariance is ill-conditioned ($cond(Q)$of 7.9×10^5 in 2016 and 680-1.900 from 2018) because its carbon price is almost constant within years. As predicted by theory, $tr(P_{t|t})$ converges to its Riccati value in one to six months. Since the state model feeds only the innovations used in the climate betas, and the ablation $\kappa = 0$ has small effects (Section 4.8), these aspects do not drive the results.

4.6 Out-of-Sample Performance

Table 7 reports net-of-cost performance over 2018–2025, and Figure 3 the cumulative wealth. Four regularities emerge, which are descriptive until tested in Section 4.7. First, the period was adverse for Latin American equities in USD: volatilities are 26-33%, the pandemic crash dominates the drawdowns, and Chile had near-zero returns with drawdowns of about 60%. Second, no strategy dominates. In Sharpe ratio, $1/N$-restr is the best in Brazil, Mexico and Chile. In Colombia DRO-Eucl (0.316) is followed by $1/N$-restr (0.308) and DRO-CC (0.305). In CVaR, DRO-CC ranks second among the five optimized strategies in Mexico and Chile and third in Brazil and Colombia, and it has the smallest drawdown of all strategies in Brazil and Mexico.

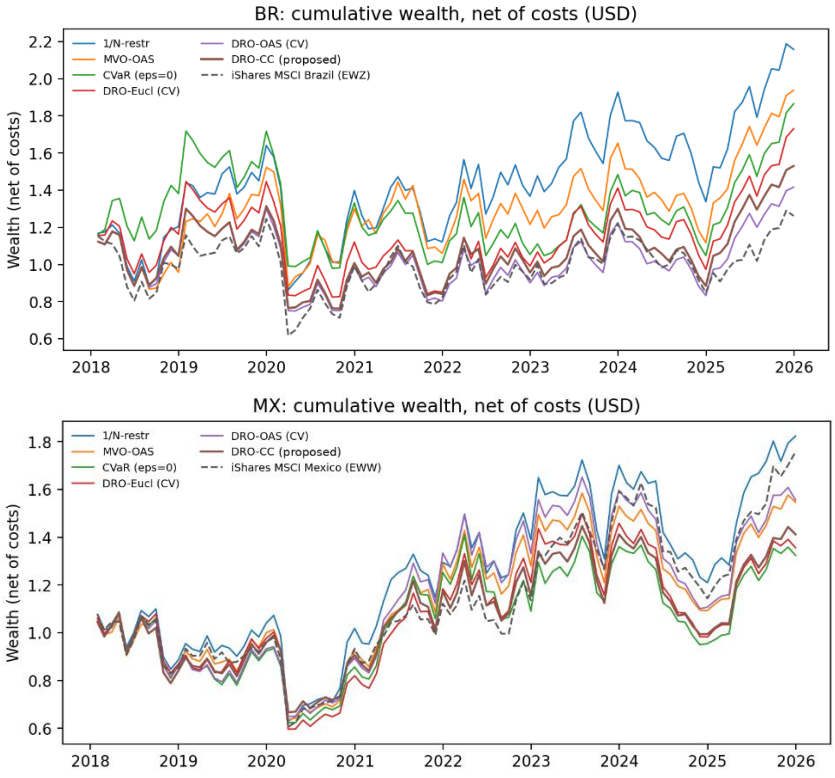
Table 7. Out-of-sample performance, Jan-2018-Dec-2025 (96 months, USD, net of costs)

Market	Strategy	Ret. (%)	Vol. (%)	Sharpe	$CVaR_{5\%}$	MaxDD	Turnover
Brazil	1/N-restr	10.09	32.83	0.387	19.45	−47.40	0.18
	MVO-OAS	8.62	31.51	0.343	18.74	−42.18	0.25
	CVaR (ε = 0)	8.11	30.77	0.327	17.02	−43.00	0.24
	DRO-Eucl (CV)	7.10	30.94	0.297	17.47	−43.00	0.29
	DRO-OAS (CV)	4.45	30.79	0.215	18.09	−42.43	0.30
	DRO-CC	5.46	30.66	0.246	17.80	−41.96	0.37
	DRO-KF	9.71	32.54	0.377	19.22	−46.88	-
	ETF (EWZ)	2.95	32.42	0.181	19.74	−50.35	-
Mexico	1/N-restr	7.80	28.40	0.325	18.48	−42.70	0.17
	MVO-OAS	5.60	27.38	0.248	17.65	−40.04	0.24
	CVaR (ε = 0)	3.58	27.53	0.176	17.45	−42.74	0.30
	DRO-Eucl (CV)	3.88	28.40	0.194	18.57	−44.74	0.46
	DRO-OAS (CV)	5.68	27.10	0.247	16.41	−39.66	0.34
	DRO-CC	4.41	26.61	0.203	16.77	−38.45	0.37
	DRO-KF	7.41	27.95	0.312	18.27	−42.49	-
	ETF (EWW)	7.30	25.28	0.312	17.06	−43.78	-
Colombia	1/N-restr	7.28	26.61	0.308	17.00	−42.05	0.14
	MVO-OAS	6.04	26.10	0.265	16.53	−44.85	0.24
	CVaR (ε = 0)	6.28	26.47	0.274	17.95	−47.10	0.22
	DRO-Eucl (CV)	7.46	26.40	0.316	16.96	−45.76	0.18

	DRO-OAS (CV)	6.38	26.27	0.277	16.72	−46.23	0.26
	DRO-CC	7.14	26.11	0.305	16.94	−43.43	0.29
	DRO-KF	7.30	26.43	0.308	16.95	−41.62	-
	ETF (ICOLCAP)	4.92	27.30	0.228	17.33	−46.50	-
Chile	1/N-restr	3.08	27.83	0.158	15.66	−59.26	0.09
	MVO-OAS	0.32	27.87	0.062	17.58	−60.78	0.16
	CVaR (ε = 0)	0.99	27.95	0.086	16.50	−59.02	0.19
	DRO-Eucl (CV)	0.92	28.18	0.086	17.24	−59.61	0.25
	DRO-OAS (CV)	0.20	28.02	0.059	17.59	−61.19	0.20
	DRO-CC	0.34	27.52	0.060	16.67	−59.71	0.29
	DRO-KF	2.52	27.65	0.137	15.79	−59.30	-
	ETF (EWZ)	0.27	27.17	0.053	15.33	−59.62	-

Notes: Ret.: annual return. Vol.: annualized volatility. Sharpe: annualized, in excess of the T-bill. CVaR5%: mean of the monthly returns in the 5% tail, sign changed. MaxDD: maximum drawdown. Turnover: mean $\| w_t - w_{t-} \|_1$ per rebalancing.

Source: own elaboration with data from Yahoo Finance.



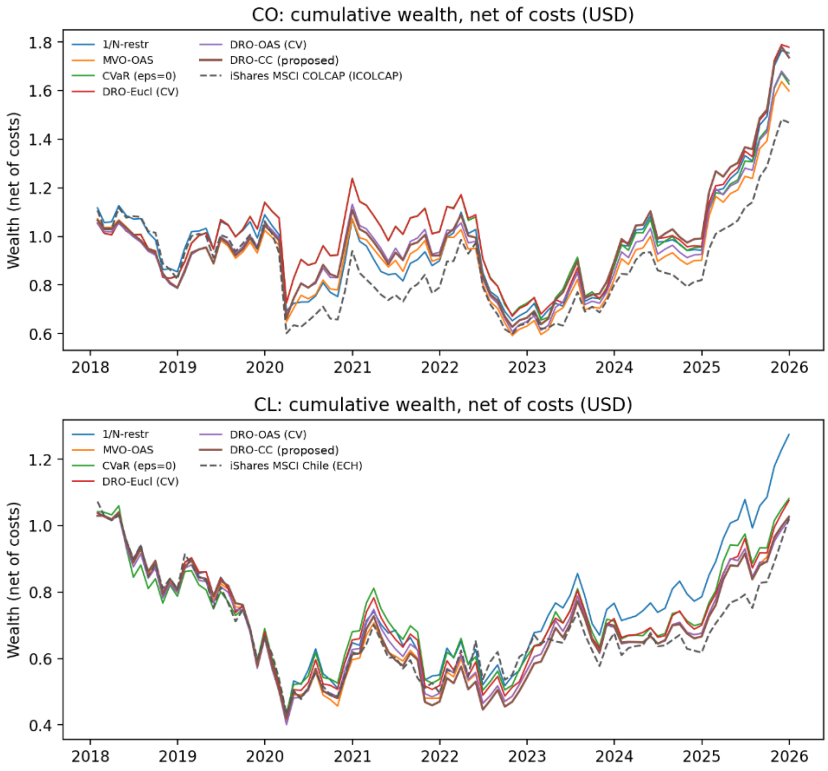


Figure 3. Cumulative wealth net of costs (USD), January 2018–December 2025.

Source: own elaboration with data from Yahoo Finance.

Third, DRO-CC and its climate-free counterpart behave similarly: DRO-CC has a higher return than DRO-OAS in Brazil, Colombia and Chile and a lower one in Mexico, with CVaR differences below one point. DRO-CC is among the most concentrated and most active strategies, a consequence of operating near the minimum-variance regime. Fourth, the transition-aligned portfolios have tail risk comparable to the country ETFs: DRO-CC has a lower CVaR than the ETF in Brazil, Mexico and Colombia and a higher one in Chile, and a higher Sharpe ratio in Brazil, Colombia and Chile.

The specification of the submitted version (DRO-KF) is almost identical to $1/N$-restr in every market (with nearly equal returns and CVaR). By Proposition 1 with M = I, its penalty $\varepsilon\| w \|_2$ omits the factor $\alpha^{-1/2}$, so its effective radius is $\sqrt{\alpha}\,\varepsilon = 0.23 - 0.34$, and the Euclidean norm is minimized by equal weights. The favorable results of the submitted version in Chile and Colombia over 2023–2025 therefore reflected naïve diversification under the constraints, not climate information; in addition, an implementation error assigned $1/N$ weights whenever the solver failed.

4.7 Statistical Comparison

Table 8 reports the comparisons of DRO-CC with each benchmark by market (Panel A) and pooled across markets (Panel B). No difference survives the Holm adjustment. Without adjustment, the smallest p-values are the Sharpe difference against DRO-OAS in Brazil (+0.031, one-sided $p = 0.065$), the Sharpe difference against $1/N$-restr in Brazil (−0.141,

two-sided p = 0.066) and the DM test against CVaR ($\varepsilon = 0$) in Colombia ($p = 0.049$). With a fixed 12-month block, the Brazilian Sharpe interval against DROOAS excludes zero ($p = 0.032$), which does not survive the adjustment across four markets; with blocks of 3 and 6 months and with the automatic rule the conclusions are unchanged.

The panel interval against DRO-OAS has a half-width of about half a point of monthly CVaR, around 3% of the level of CVaR (16-18%). This is a precise null: the data rule out that the climate component moves pooled monthly CVaR by more than about 0.5 points in either direction. The intervals against MVO-OAS and $1/N$-restr are wider and centered on small negative values, which is compatible with a modest benefit of the robust model but does not establish it.

In light of this evidence, H1 is not supported: the one-sided p-values against DRO-OAS range from 0.15 to 0.70 and the pooled difference is -0.04 points. H2 is not supported: the point estimates against MVO-OAS are negative in Brazil, Mexico and Chile and positive in Colombia, and against $1/N$-restr negative in three markets, but no interval excludes zero. H3 is not supported: of the two markets with the highest measured physical exposure, the climate component lowers CVaR in Chile (−0.92 points) and raises it in Mexico (+0.36 points), and neither difference is significant.

Table 7. DRO-CC against the benchmarks: bootstrap intervals and DM tests (2018–2025)

Panel A. By market

Market	DRO-CC vs.	ΔCVaR	95% CI	p1	ΔSharpe	95% CI	DM-HLN	P2
Brazil	DRO-OAS (CV)	−0.29	[−1.49; 0.69]	0.175	0.031	[−0.010; 0.076]	−1.59	0.114
	MVO-OAS	−0.93	[−2.49; 0.83]	0.119	−0.097	[−0.272; 0.058]	−1.04	0.302
	CVaR (ε = 0)	0.78	[−1.57; 3.44]	0.639	−0.081	[−0.286; 0.095]	0.34	0.733
	1/N-restr	−1.64	[−5.34; 1.29]	0.253	−0.141	[−0.329; 0.009]	−0.89	0.376
	Country ETF	−1.94	[−5.50; 1.39]	0.164	0.065	[−0.189; 0.310]	−1.17	0.245
Mexico	DRO-OAS (CV)	0.36	[−0.48; 1.82]	0.700	−0.044	[−0.140; 0.073]	0.86	0.389
	MVO-OAS	−0.88	[−2.50; 0.46]	0.185	−0.045	[−0.173; 0.064]	−0.52	0.606
	CVaR (ε = 0)	−0.68	[−1.91; 0.61]	0.172	0.028	[−0.103; 0.180]	−1.34	0.183
	1/N-restr	−1.71	[−5.41; 1.22]	0.224	−0.122	[−0.334; 0.060]	−0.67	0.503
	Country ETF	−0.28	[−2.80; 2.68]	0.506	−0.109	[−0.339; 0.084]	0.50	0.615
Colombia	DRO-OAS (CV)	0.23	[−1.53; 2.06]	0.698	0.028	[−0.026; 0.099]	−0.14	0.891
	MVO-OAS	0.41	[−1.07; 1.73]	0.840	0.040	[−0.036; 0.125]	0.23	0.815
	CVaR (ε = 0)	−1.01	[−3.12; 1.16]	0.110	0.031	[−0.088; 0.153]	−2.00	0.049
	1/N-restr	−0.05	[−2.54; 2.62]	0.523	−0.002	[−0.198; 0.208]	0.22	0.823
	Country ETF	−0.39	[−4.35; 3.65]	0.376	0.077	[−0.207; 0.332]	−1.08	0.285
Chile	DRO-OAS (CV)	−0.92	[−2.04; 0.56]	0.148	0.001	[−0.072; 0.075]	−0.98	0.331
	MVO-OAS	−0.91	[−2.31; 1.01]	0.277	−0.002	[−0.056; 0.053]	−0.31	0.756
	CVaR (ε = 0)	0.17	[−0.88; 1.11]	0.586	−0.026	[−0.141; 0.089]	−0.06	0.950
	1/N-restr	1.01	[−1.81; 4.16]	0.705	−0.098	[−0.255; 0.052]	0.52	0.605
	Country ETF	1.35	[−1.87; 5.75]	0.746	0.006	[−0.272; 0.323]	0.51	0.609

Panel B. Pooled across markets (384 stacked observations)

DRO-CC vs.	ΔCVaR	95% CI	P2	Block
DRO-CC vs. DRO-OAS (CV)	10.09	32.83	0.387	19.45
DRO-CC vs. MVO-OAS	8.62	31.51	0.343	18.74
DRO-CC vs. 1/N-restr	4.45	30.79	0.215	18.09
DRO-CC vs. DRO-OAS, design 2022–2025	9.71	32.54	0.377	19.22

Note: Δ: DRO-CC minus benchmark; negative ΔCVaR and positive ΔSharpe favor DRO-CC. CVaR in percentage points of monthly return. CI: stationary bootstrap percentile interval (2,000 replications, automatic block length). p1: one-sided bootstrap p-value for ΔCVaR< 0. DM-HLN: Diebold–Mariano statistic with HLN correction on $L(r) = (minr, 0)2$; p2: two-sided p-value. All Holm-adjusted p-values across markets equal 1, except DM against CVaR ($\varepsilon = 0$) in Colombia (0.98).

Source: own elaboration.

4.8 Contribution of the Climate Component and Robustness

Table 9 isolates the two climate channels by removing them one at a time from DRO-CC with the same radius grid.

Table 9. Effect of the climate components on DRO-CC: specification with the component minus specification without it

	Kernel and metric		**Kernel only (ω)**		**Metric only (κ)**	
	ΔRet	**ΔCVaR**	**ΔRet**	**ΔCVaR**	**ΔRet**	**ΔCVaR**
Brazil	−0.43	−0.05	−0.07	−0.06	+0.20	+0.01
Mexico	−1.52	+0.44	−1.52	+0.44	+0.62	−0.29
Colombia	+1.10	+0.34	+0.17	+0.06	+0.68	+0.25
Chile	−0.08	−0.45	+0.90	−0.48	+0.30	+0.02

Note: ΔRet in points of annual return; ΔCVaR in points of monthly CVaR at 5% (negative favors the climate component). Ablations use a coarse radius grid for computational reasons, so the base differs slightly from Table 7.

Source: own elaboration.

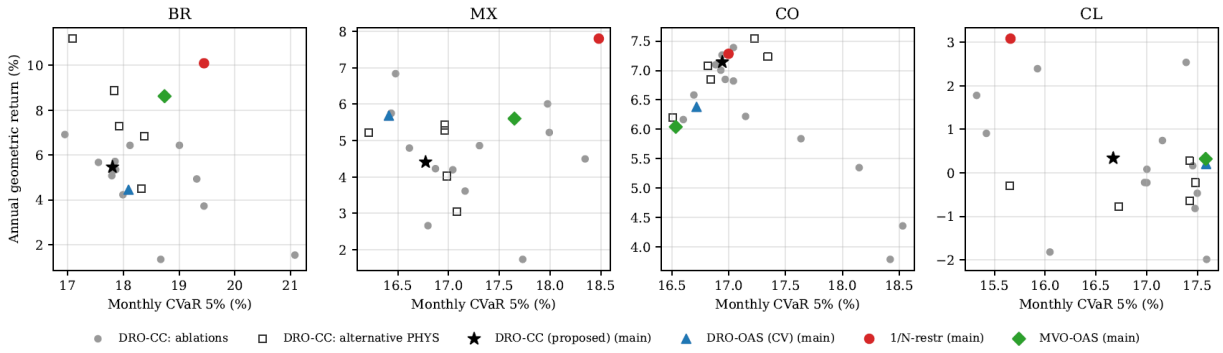


Figure 4. Annual return against monthly CVaR at 5% of DRO-CC under all ablations and alternative measurements of PHYS, with the main strategies as reference.

Source: own elaboration.

The climate components change the result in opposite directions across markets, between −1.5 and +1.1 points of annual return and between −0.45 and +0.44 points of monthly CVaR; the kernel dominates in Mexico and Chile and the metric block in Colombia. None of these differences is stable in sign, and all are smaller than the variation produced by design choices without climate content (Figure 4). In Brazil, the return of DRO-CC ranges from 1.3% (36-month rolling library) to 11.2% (PHYS without market control) and its CVaR from 16.9% to 21.1%; the largest sources of variation are the scenario library, the kernel bandwidth (Mexico: 1.7% with h = 0.5 against 4.2% in the coarse-grid base), the position cap and the measurement of PHYS. Alternative PHYS measurements move the Brazilian return from 4.5% (headquarters rule) to 6.8% (water stress only), 7.3% (six lags) and 11.2% (no market control); the last figure should not be read as an improvement, because it would be chosen after observing the result. Tightening the physical constraint by 30% worsens performance in Brazil (return of 1.5%, CVaR of 21.1%), which quantifies the cost of requiring lower physical exposure with the current measure.

Two robustness checks lead to the same conclusion. In the design estimated up to 2021 and evaluated over 2022–2025, with PHYS fixed (one of 55 issuers with a significant ENSO response), the pooled difference against DRO-OAS is +0.09 points ($p = 0.60$). The only difference that survives the Holm adjustment is against CVaR ($\varepsilon = 0$) in Mexico (ΔCVaR of −1.74 points, adjusted $p < 0.001$; ΔSharpe of +0.17, adjusted $p = 0.10$), which shows the value of robustness against the empirical CVaR, not of climate information. In the main design restricted to 2022–2025, 1/N-restr has the highest Sharpe ratio in Mexico, Colombia and Chile and DRO-Eucl in Brazil, and DRO-CC ranks second in Colombia and Chile. Finally, the replication of the radius of the submitted version explains its behavior. That radius depended for 96–98% on the Mahalanobis distance between the filtered state and the IS mean, and for less than 0.6% on the trace of the posterior covariance; it therefore measured the drift of trending variables rather than uncertainty. The bounded logit transformation was undefined for Brazil's zero carbon price in all months and for Colombia and Chile before 2017, and the late peak of the Chilean radius coincides with six OOS months in which copper exceeded its IS range.

4.9. Discussion

The results are consistent with three strands of literature. First, the convergence of the submitted specification to 1/N-restr is the mechanism described by Pflug et al. (2012): with a Euclidean transport cost and a large radius, robustness is equivalent to diversification, and 1/N is hard to beat out of sample (DeMiguel et al., 2009). With a covariance-based metric, the robust portfolio instead approaches minimum variance under the metric, in line with the equivalence between Wasserstein robustness and regularization (Blanchet et al., 2022); the geometry of the ambiguity set, not only its size, determines what robustness means. Second, the weak identification of the radius illustrates the practical difficulty of selecting it by

validation (Mohajerin Esfahani and Kuhn, 2018; Gao, 2023): with 96 months and portfolios that differ little, the signal-to-noise ratio of the criterion is low.

Third, the evidence on the ENSO channel qualifies the motivation of climate-aware allocation in the region. ENSO has persistent macroeconomic effects (Cashin et al., 2017; Callahan and Mankin, 2023) and the market response to its news is detectable in 2014–2017, but it is common to the market and episodic, so it cannot be exploited by constraints on relative physical exposure. This complements the underreaction to drought documented by Hong et al. (2019) and the hedging of aggregate climate news with portfolios (Engle et al., 2020): in Latin American equities, ENSO behaves as a systematic factor more than as a source of cross-sectional dispersion.

The heterogeneity of physical shocks at the asset level (Bressan et al., 2024) suggests that capturing differential exposure requires asset-level operational data (for instance, hydropower dependence) rather than return-based sensitivities. Regarding transition risk, the constraints aligned with the EU benchmarks did not impose a visible cost relative to the country ETFs, which is consistent with a modest carbon premium over this period (Bolton and Kacperczyk, 2023; Pástor et al., 2021), although our design does not identify it. For the regional literature (Roncoroni et al., 2021; Pesce et al., 2024; Monzón Citalán et al., 2025), the contribution is an OOS, cost-adjusted evaluation of climate-informed allocation with explicit inference, whose null result is informative about the limits of current public climate data for portfolio decisions.

5. Conclusions and Policy Implications

This paper develops and evaluates a distributionally robust portfolio model in which climate information shapes the ambiguity set: a filtered climate state reweights historical scenarios and enters the transport cost of a type-2Wasserstein ball, and the exact worst-case CVaR reduces to a second-order cone program whose penalty is the portfolio volatility under a climate-augmented metric. The model is evaluated out of sample in Brazil, Mexico, Colombia and Chile over 2018–2025, with transition limits aligned with the EU Paris-aligned benchmarks, a data-based physical exposure, transaction costs and nested benchmarks.

The main result is a precise null. The climate-conditioned model does not differ detectably from its climate-free counterpart: the pooled difference in monthly CVaR is −0.04 points, with a 95% interval of [−0.54, 0.43], and no market-level difference survives the Holm adjustment. The effect of climate components changes sign across markets and is smaller than the effect of design choices without climate content. Three findings explain this result: equity responses to ENSO news are mostly market-wide and concentrated in 2014–2017, so physical-exposure constraints across firms have little to exploit; the radius is weakly identified by cross-validation, and the model operates mostly near the minimum-variance regime, where the kernel loses influence; and, with a Euclidean metric, distributional robustness reduces to diversification, which explains the results of the submitted version.

For practice, the implications are proportional to the evidence. Transition limits aligned with the EU benchmarks can be met with tail risk comparable to the country ETFs, and the closed form of Proposition 1 makes climate-conditioned robustness tractable and auditable. However, we do not find that conditioning on public climate information improves tail-risk control, and investors should not expect that return-based physical-exposure measures capture material firm-level differences. The results do not support using this framework as a supervisory stress-testing tool.

The study has limitations. The evaluation spans a single complete ENSO cycle; transition intensities are expert-based priors; the universe was selected with 2020 capitalization; temperature anomalies are regional averages that could be replaced by higher-resolution re-analyses such as ERA5 (Hersbach et al., 2020) or CRU TS (Harris et al., 2020); and the state-space model is ill-conditioned for Mexico. Natural extensions are asset-level physical-exposure data, such as hydropower dependence or the location of mining operations, a joint multi-market ambiguity set in a common currency, and a longer evaluation that includes earlier ENSO events.

Data and code availability

All series come from the public sources listed in Table 2. The replication notebook, which implements every result reported in the paper and the replication of the submitted specification, is available from the author and will be deposited in a public repository in GitHub **https://github.com/climrisklatam1/ClimRiskLA/**

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